

# Markov chains with stationary distributions



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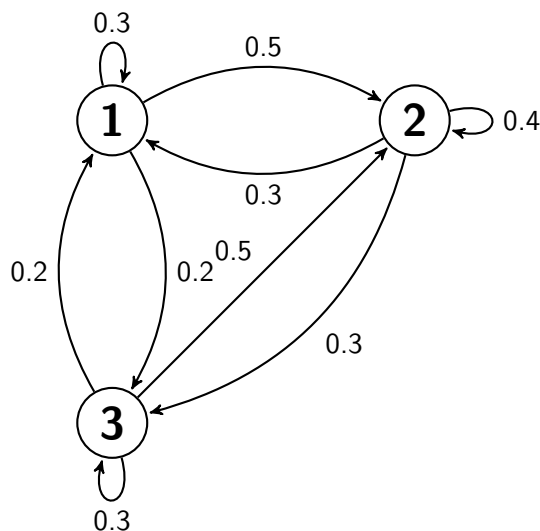
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## Abstract

Numerical simulations of an aperiodic Markov chain with all recurrent states have been performed. The limiting behaviour of the chain is computed by sampling and by using the Chapman Kolmogorov relationship and the results obtained using these two methods are shown to be the same. The fraction of time that the system would expect to spend in each of the states of the chain is then computed by sampling. This distribution is found to be the same as the limiting stationary distribution that is found by computing the principle (left) eigenvector of the transition matrix.

## 1 Introduction



**Fig. 1.** The transition graph that was simulated for this report.

I chose to simulate the Markov chain with the transition graph shown in figure 1 for my project. The transition matrix for this graph is as follows:

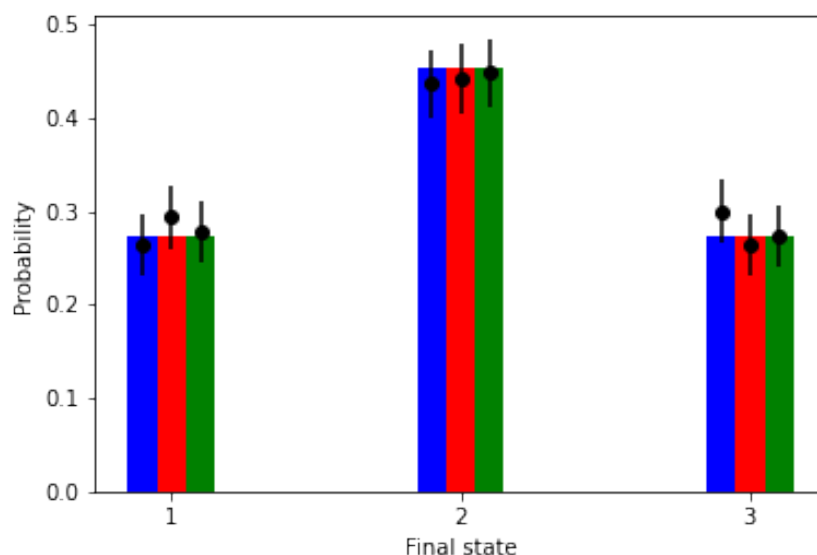
$$\mathbf{A} = \begin{pmatrix} 0.3 & 0.5 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}$$

All the states in this chain are recurrent and aperiodic. The chain thus has a limiting stationary distribution, which I will investigate in this report. In section 2 I will investigate

the limiting behaviour of the chain by simulating the chain and by using the Chapman-Kolmogorov equation. In section 3 I will investigate the probabilities of being in each of the states by performing simulations and by calculating the principle (left) eigenvector of the transition matrix.

## 2 Limiting Behaviour

To investigate the limiting behaviour of the Markov chain I ran a series of simulations. In these simulations the chain was propagated for 100 steps and the final state the system arrived in was recorded. This procedure was repeated 500 times and the black dots in figure 2 show the histograms that were constructed from the end state for these 500 simulations. Three separate histograms were constructed as separate sets of simulations were run starting from each of the three initial states. The error bar on these points indicate the 90% confidence limit on the estimate of the probability.



**Fig. 2.** The probabilities of finishing in each of the three states when the a Markov chain illustrated in figure 1 is run forward for 100 steps. The blue, red and green bars indicate the conditional probabilities of finishing in each of the state if the chain starts in states 1, 2 and 3 respectively. The black error bars were calculated by running 500 simulations of the chain and computing a histogram. The error bars on these estimates represent the 90% confidence limit. The heights of the bars are then the elements of the 100 step transition matrix, which was calculated using the Chapman-Kolmogorov relationship.

Figure 2 shows that the conditional probability of finishing in each of the states in the chain does not have a statistically significant dependence on the initial state. The conditional probability of being in state 1 after 100 steps given you started in state 1 is the same as the conditional probability of being in that state given you started in state 2 or 3.

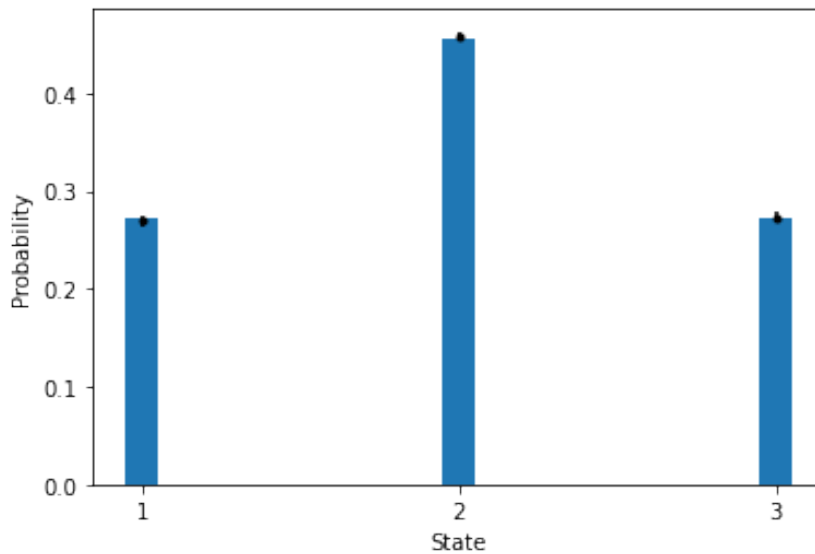
The conditional probabilities of being in the various states can be calculated using the Chapman-Kolmogorov relationship, which states:

$$\mathbf{A}^{n+m} = \mathbf{A}^n \mathbf{A}^m$$

In this expression  $\mathbf{A}^n$  is the  $n$  step transition probability matrix. We can thus use the expression above to calculate the probability of being in each of the states after 100 steps by taking the hundredth power of the 1-step transition probability matrix that was given in the previous section. The elements of this matrix are illustrated using the coloured bars in figure 2. You can see that the transition probabilities that we obtain this way are within the statistical errors on the estimates of these probabilities that we obtained by sampling.

### 3 Stationary distribution

We wanted to investigate the fraction of time that the Markov chain spends in each of the various states. To do this we ran 24000 steps of the chain starting from state 1 and constructed a histogram that measured the number of times each state was visited. This histogram is indicated by the black dots in figure 3. In calculating errors on this estimate of the histogram we used the block averaging technique with blocks of length 400 steps. Using this technique ensures that the variance is not underestimated because of the correlations between the values of the random variable at adjacent times. Error bars that represent a 90% limit are shown in figure 3 but these errors are not visible as they are narrower than the widths of the symbols that have been used to indicate the value of the probabilities.



**Fig. 3.** The stationary distribution for the Markov chain that has been studied in this work. The black points indicate the estimate of this distribution that was obtained by sampling the chain as well as the 90% confidence limit on this estimate of the probabilities. The heights of the bars are the elements of the top left-eigenvector of the transition matrix.

All the states of the Markov chain are recurrent and aperiodic. We can thus find the limiting stationary distribution by finding the top (left) eigenvalue of the transition matrix. The blue bars in figure 3 indicate the elements of this eigenvector. You can clearly see that the stationary distribution we get by finding the top eigenvector falls within the error bars on the estimate of the stationary distribution that we get by sampling.